

DIVISION 9 - FINISHES
Section 09830 Elastomeric Coatings

Part 1 – General

1.01 Summary

- A. This specification describes the coating of substrates with an elastomeric, crack bridging, anti-carbonation, protective coating.

1.02 Quality Assurance

- A. Manufacturing qualifications: The manufacturer of the specified product shall be ISO 9001:2008 certified and have in existence a recognized ongoing quality assurance independently audited on a regular basis.
- B. Contractor qualifications: Contractor shall be qualified in the field of concrete repair and protection with a successful track record of 5 years or more. Contractor shall maintain qualified personnel who have received product training by a manufacturer's representative.
- C. Install materials in accordance with all safety and weather conditions required by manufacturer or as modified by applicable rules and regulations of local, state and federal authorities having jurisdiction. Consult Material Safety Data Sheets for complete handling recommendations.

1.03 Delivery, Storage, and Handling

- A. All materials must be delivered in original, unopened containers with the manufacturer's name, labels, product identification, and batch numbers. Damaged material must be removed from the site immediately.
- B. Store all materials off the ground and protect from rain, freezing or excessive heat until ready for use.
- C. Condition the specified product as recommended by the manufacturer.

1.04 Job Conditions

- A. Environmental Conditions: Do not apply material if it is raining or snowing or if such conditions appear to be imminent. Minimum application temperature 7°C and rising.

- B. Protection: Precautions should be taken to avoid damage to any surface near the work zone due to mixing and handling of the specified material.

1.05 Submittals

- A. Submit two copies of manufacturer's literature, to include: Product Data Sheets, and appropriate Material Safety Data Sheets (MSDS).

1.06 Warranty

- A. Provide a written warranty from the manufacturer against defects of materials for a period of one (1) year, beginning with date of substantial completion of the project.

Part 2 – PRODUCTS

2.01 MANUFACTURER

- A. Vetotop AC441 as manufactured by Saudi Vetonit Company Ltd.-Saveto
71 St. Second Industrial Estate Riyadh P.O.Box 52235 Riyadh 11563 Saudi Arabia Tel +966 11 2653334, Fax: +966 11 265 0936, E-mail: info.Riyadh@saveto.com; Web: www.saveto.com.
- B. Vetoprime AP443 as manufactured by Saudi Vetonit Company Ltd.-Saveto
71 St. Second Industrial Estate Riyadh P.O.Box 52235 Riyadh 11563 Saudi Arabia Tel +966 11 2653334, Fax: +966 11 265 0936, E-mail: info.Riyadh@saveto.com; Web: www.saveto.com

2.2 MATERIALS

- A. Acrylic protective and decorative coating for concrete and masonry.
 - 1. Vetotop AC441.
 - 2. Water-based acrylic protective and decorative coating for concrete and masonry.
- B. Primer for use with Vetotop AC441 anti carbonation top coat
 - 1. Vetoprime AP443
 - 2. Water repelling & anti chloride primer for use with Vetotop AC441 anti carbonation top coat

2.3 MATERIALS PROPERTIES

A. Vetotop AC441

1. Volume Solids: 50% approx.
2. Elongation at Break (ASTM D412): > 300%.
3. Carbon Dioxide Diffusion Resistance (AS/NZS 4548.5-1999)
4. Equivalent thickness of air (R), m :> 96 meters
5. Equivalent thickness of 30 MPa concrete cover (Sc):>240 mm
6. CO₂ Diffusion Coefficient, cm² / sec: 3.7 x 10⁻⁷
7. Water vapour transmission resistance (AS/NZS 4548.5-1999):
8. Vapour Transmission Rate: 55.7g/m²/24 hours
9. Equivalent thickness of Air (Sd) m :< 1
10. Vapour Diffusion Coefficient cm²/sec: 1.1x10⁻⁴
11. Chloride ion diffusion coefficient @ 147 days: Chloride ion diffusion coefficient @ 147 days
12. Incipient crack spanning capability @ 200 microns dft; After 2000 hours accelerated weathering:2mm
13. Water Transmission Resistance: 14g/24h/m²/kPa
14. Fire testing Spread of flame: Class 1

B. Vetoprime AP443

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|-----------------------------|--------------|
| 1. Solid Content: | %7 |
| 2. Wet Appearance: | Clear |
| 3. Dry Appearance: | Transparent. |
| 4. Dry Time (25°C & RH50%): | 25 minutes |

Part 3 – Execution

3.01 Surface Preparation

- A. All surfaces should be dry and free from any contamination and all traces of mould release oils and curing compounds by lightly grit-blasting the surface.
- B. Where moss, algae or similar growths have occurred, treatment with a proprietary biocide should be carried out after the grit-blasting process.
- C. Surfaces containing blow-holes or areas of pitting should first be filled using Vetorep CR523 a cementitious fairing coat. Rougher substrates can be levelled using a cementitious product from Saveto Concrete Repair Range.

3.02 Mixing and Application

- A. Mixing: Stir materials to ensure uniformity using a low speed (400-600 rpm) drill and paddle. To minimize color variation, blend two batches of material.

- B. Crack detail: Recommended application temperatures 4-38 °C. Small defects and cracks (non-structural): Cracks 10 – 20 mils. Apply Surface Filler "Brush Grade" generously over the center of the cracks. Feather material to zero over a two-inch wide area. Allow a minimum 24 hours to cure before over coating. Large defects and cracks (non-structural): Cracks >20mils. Rout to 1/4-in wide by 1/4-in. deep. Blow out cut with oil-free compressed air. Fill slot with Surface Filler "Knife Grade" allowing for a small crest to remain. This will compensate for any shrinkage that might occur. Allow 24 hours-minimum cure before over coating.
- C. Coating Application: Apply by brush, roller, or spray over entire area moving in one direction. A minimum of two **C**oats are required. Each coat should be applied at a rate of 0.4 liter/sqm. Allow a minimum of 2 hours prior to re-coating.
- D. Adhere to all limitations and cautions for the elastomeric acrylic coating in the manufacturers printed literature.

3.03 Cleaning

- A. Vetotop AC441 should be removed from tools and equipment with clean water immediately after use.
- B. The cured elastomeric acrylic coating can be removed mechanically.
- C. Leave finished work and work area in a neat, clean condition without evidence of spillovers onto adjacent areas.